

Effects of Medicaid Interstate Drug Purchasing Pools on Supplemental Rebate Capture

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March 2026

Abstract

Objective: To estimate the causal effect of interstate Medicaid drug purchasing pool participation on states' supplemental rebate performance; specifically, whether joining the Sovereign States Drug Consortium (SSDC) increases annual supplemental rebate percentages relative to alternative pool membership (NMPI, TOP\$) or no pool participation.

Data Sources: CMS-64 Financial Management Reports, validated against CMS State Drug Utilization Data, state-level, 2002–2024; a new timeline of pool participation and concurrent policy changes built from web scraping, direct state outreach, FOIA requests, and expert interviews.

Study Design: Staggered-adoption difference-in-differences, event study models; state Medicaid programs observed annually, 2002–2024; states switching to SSDC compared against other-pool (TOP\$, NMPI) and no-pool states; robustness checks include parallel trends assessments and alternative control group definitions.

Principal Findings: States that joined SSDC saw large gains in supplemental rebate capture. In the preferred specifications – SSDC versus all non-SSDC states from 2010 onward (N=720) and SSDC versus no-pool states over the full panel (N=735) – SSDC participation increased supplemental rebate capture by around 3.1 percentage points annually, with effects reaching 4.8–5.7 percentage points at longer post-adoption horizons, and 8 of 11 post-adoption coefficients

were significant at the 5% level in each specification. Event studies supported parallel pre-trends in both preferred specifications. The gains arrived through the SSDC rather than through pooling generally: a direct comparison of SSDC to NMPI/TOP\$ states showed significant pre-trend violations, which precludes a causal interpretation of that comparison but suggests that SSDC and other-pool states were on fundamentally different trajectories before SSDC adoption. Interstate purchasing pool participation is therefore not uniformly effective in increasing supplemental rebates, and features beyond pooled population size, such as negotiating approach and administrative methodology, may be the key drivers. States aiming to increase Medicaid supplemental rebates should not assume that pool participation alone will yield gains; broader adoption of SSDC methods could generate substantial system-wide savings, potentially over \$1 billion annually, and improved transparency around pool participation and related policy changes would support more rigorous evaluation and better-informed purchasing decisions.

1. Introduction

Medicaid is the single largest payer for prescription drugs in the United States. In fiscal year 2023, gross Medicaid drug spending was over \$80 billion, and net spending (after rebates) reached \$51 billion and continues to rise, driven by high-cost specialty and biologic therapies (MACPAC, 2022). As such, managing pharmaceutical expenditures is a first-order fiscal challenge for state Medicaid programs.

The Medicaid Drug Rebate Program (MDRP), established by the Omnibus Budget Reconciliation Act of 1990, requires manufacturers to pay statutory rebates to Medicaid in exchange for coverage of their products. These federal rebates are substantial: the minimum rebate for brand-name drugs is 23.1% of Average Manufacturer Price (AMP) or the AMP-best-

price difference, whichever is greater, plus an inflation penalty (KFF, 2023). However, states also have the option to negotiate supplemental rebates with manufacturers above and beyond the statutory minimum, typically in exchange for favorable placement on the state's Preferred Drug List (PDL). As of 2019, 47 states and the District of Columbia had supplemental rebate agreements, which collectively generated over \$1.7 billion annually (OIG, 2014; KFF/HMA, 2020).

To increase their bargaining leverage in these negotiations, many states have joined interstate drug purchasing pools, which are multi-state coalitions that negotiate supplemental rebates collectively. Three major pools emerged in the mid-2000s: the National Medicaid Pooling Initiative (NMPI, established 2003), The Optimal PDL Solution (TOP\$, established 2005), and the Sovereign States Drug Consortium (SSDC, established 2005). By 2024, 33 of 50 states participated in one of these three pools. However, despite their widespread adoption, there is no causal evidence on whether these pools actually increase supplemental rebate capture, or on whether all pools are equally effective.

This question has important policy implications. Descriptive data suggest large variation in supplemental rebate performance across states: in 2024, supplemental rebates ranged from near zero in some states to over 28% of drug spending in others. The six states with the highest median rebate performance over 2018–2024 (Delaware, Oklahoma, Vermont, Wyoming, West Virginia, and Maine) share two characteristics: all participate in SSDC, and all have uniform PDLs that apply across both fee-for-service and managed care. Meanwhile, NMPI and TOP\$ member states have rebate performance that is largely indistinguishable from non-pool states. If these patterns reflect causal effects of pool membership rather than selection, the fiscal

implications are significant – even modest increases in supplemental rebate capture, scaled across all 50 states, represent billions of dollars in potential savings.

Estimating the causal effect of pool membership is challenging, however. States that join pools may differ systematically from those that do not in ways that also affect rebate performance, including Medicaid program size, pharmacy benefit structure (fee-for-service vs. managed care, carve-in vs. carve-out), PDL design and enforcement, and concurrent policy changes. In addition, pool membership decisions are often intertwined with simultaneous changes to PDL vendors, managed care contracts, and drug utilization review processes. This creates attribution problems that descriptive comparisons cannot resolve.

This paper addresses these challenges by constructing a new state-level panel dataset spanning 2002–2024 and applying staggered-adoption difference-in-differences methods. The outcome variable, the annual supplemental rebate percentage, is calculated from CMS-64 Financial Management Reports and validated against State Drug Utilization Data. To address the attribution problem, I constructed a detailed timeline of pool participation and concurrent policy changes for all 50 states using web scraping of state Medicaid websites, direct outreach to state pharmacy officials, and Freedom of Information Act (FOIA) requests. The identification strategy exploits the staggered timing of SSDC adoption across 15 states between 2006 and 2024, using never-treated and other-pool states as comparison groups, with event-study specifications that accommodate heterogeneous treatment effects across adoption cohorts.

The analysis produces three main findings. First, in the preferred specifications, SSDC participation increases supplemental rebate capture by around 3.1 percentage points annually on average, with effects growing to 4.8–5.7 percentage points at longer post-adoption horizons. Second, a direct comparison of SSDC to NMPI/TOP\$ states is confounded by significant pre-

trend violations. SSDC adopters and other-pool states were on fundamentally different rebate trajectories before adoption, which shows how much the choice of comparison group matters in this setting. Lastly, restricting the sample to 2010 onward, after the pools had matured, produces the strongest and most precisely estimated effects. This is consistent with SSDC's negotiating effectiveness improving over time. The results are robust to alternative control group definitions and parallel trends diagnostics.

This paper adds to the literature in a few ways. To this author's knowledge, it provides the first quasi-experimental evidence on the effectiveness of Medicaid interstate drug purchasing pools, a question that has not been empirically examined despite the pools' widespread adoption and the billions of dollars at stake. It also shows that pooling by itself is not sufficient to increase supplemental rebates; the pool's contracting structure, negotiating methodology, and administrative approach appear to be critical determinants. In doing so, it fills a gap at the intersection of the Medicaid pharmacy benefit literature (Dranove, Ody, and Starc, 2021; Munshi et al., 2018), the pharmaceutical procurement literature (Dubois, de Bettignies, and Sampaio, 2021; Seidman and Atun, 2022), and the growing body of applied work using heterogeneity-robust difference-in-differences methods in health policy (Wang, Hamad, and White, 2024). Lastly, the dataset constructed here, which documents the full history of pool participation, PDL control, and pharmacy benefit structure for all 50 states from 2002 to 2024, is itself a contribution, as no prior public source provides this information.

2. Background

2.1 The Medicaid Drug Rebate Program

The MDRP requires manufacturers to enter into rebate agreements with the Secretary of Health and Human Services as a condition of having their drugs covered by Medicaid. For brand-name drugs, the unit rebate amount (URA) is the greater of 23.1% of AMP or the difference between AMP and the manufacturer's best price (the lowest price offered to any commercial purchaser), plus an additional inflation penalty if prices have risen faster than the Consumer Price Index (CPI-U). For generic drugs, the minimum rebate is 13% of AMP (KFF, 2023).

This pricing structure creates well-documented incentive distortions. Duggan and Scott Morton (2006) showed that manufacturers respond to Medicaid's best-price provision by raising prices to commercial purchasers, and that a 10-percentage-point increase in Medicaid market share is associated with 7–10% higher average drug prices. The Affordable Care Act extended mandatory MDRP rebates to drugs dispensed through Medicaid managed care organizations (MCOs). This closed a prior loophole but added complexity to the rebate landscape.

2.2 Supplemental Rebates and Preferred Drug Lists

Supplemental rebates are voluntary and negotiated above and beyond the statutory minimum. Manufacturers pay supplemental rebates to ensure that their products receive preferred placement on a state's PDL, which reduces prior authorization requirements and other utilization management barriers. The magnitude of supplemental rebates depends on the negotiator's bargaining leverage, which in turn depends on the credibility of the threat to assign non-preferred status and on the availability of therapeutic alternatives.

PDLs are the mechanism through which this leverage operates. Wang and Pauly (2005) documented large behavioral responses to PDL implementation in Illinois (off-PDL prescriptions

dropped from 13.5% to 1.6%) and Louisiana. Munshi et al. (2018) found that Florida's state-mandated PDL reduced overall drug use by 6% but increased plan costs by 27%. As such, PDL effects on spending are not straightforward.

States also vary substantially in their PDL architecture. Some states maintain a "uniform" PDL that applies across both FFS and MCO enrollees, giving the state direct control over formulary placement for its entire Medicaid population. Others allow MCOs to maintain their own formularies, which fragments the state's bargaining leverage. This distinction is important because supplemental rebates are negotiated at the state level against the PDL; states with uniform PDLs can credibly commit to preferred and non-preferred placement decisions for a larger share of their covered lives.

2.3 Interstate Drug Purchasing Pools

Three major interstate purchasing pools emerged in the mid-2000s to help states negotiate supplemental rebates collectively.

National Medicaid Pooling Initiative (NMPI): Established in 2003 as the first multistate Medicaid drug purchasing pool. NMPI negotiates collective Supplemental Rebate Agreements (SRAs) on behalf of member states. Each member state then selects which drugs from the negotiated agreements to include on its own PDL. By 2024, 11 states participated in NMPI.

The Optimal PDL Solution (TOP\$): Established in 2005. Like NMPI, TOP\$ negotiates collective SRAs on behalf of its members. There is limited publicly available information on its specific contracting approach; its structure is assumed to be broadly similar to NMPI based on available documentation. By 2024, 7 states participated in TOP\$.

Sovereign States Drug Consortium (SSDC): Established in 2005. SSDC is structurally distinct from the other two pools in one critical way: member states contract individually with

manufacturers using their own state-specific SRAs, while leveraging the collective purchasing power of over 14 million covered lives. Each state retains full ownership of its contracts and complete autonomy over its PDL. By 2024, 15 states participated in SSDC.

This structural distinction may explain differential effectiveness. In NMPI and TOP\$, the pool drives the negotiation and owns the contract paper, and individual states then decide which negotiated drugs to include on their PDLs. In SSDC, the state drives the negotiation and owns the paper, with the consortium providing leverage through pooled lives and a coordinated negotiating approach. Horvath (2019) documented early savings figures – Vermont reported 4.7% additional savings in its first year of SSDC participation. However, these descriptive reports lack counterfactual comparisons and cannot isolate pool effects from concurrent policy changes.

2.4 Pharmacy Benefit Structure as a Confounder

The interaction between pharmacy benefit design and supplemental rebate capture is important context for this analysis. Over the past two decades, most states have shifted pharmacy benefits into managed care (carve-in), though a growing number have recently reversed course (carve-out). Dranove, Ody, and Starc (2021) found that full privatization of Medicaid pharmacy benefits decreased drug spending by 22.4%, with effects concentrated in states giving MCOs formulary flexibility. Bendicksen and Kesselheim (2022) documented the growing trend of pharmacy carve-outs and noted that states carving out can more directly leverage pool-negotiated supplemental rebates through FFS.

Gong and Bai (2020) highlighted an additional tension: FFS programs retain expensive branded drugs to capture large inflation rebates under the MDRP formula, while MCOs adopt generics and biosimilars at higher rates. This means that pool-negotiated PDLs, which may favor branded

drugs with large supplemental rebates over lower-cost generics, can create complex tradeoffs that differ by benefit structure.

3. Data

3.1 Supplemental Rebate Outcome

The primary outcome is the annual supplemental rebate percentage, defined as supplemental rebate dollars divided by total Medicaid drug expenditures for each state-year. This measure captures the share of drug spending recovered through state-negotiated (as opposed to federally mandated) rebates.

I calculated this measure using CMS-64 Financial Management Reports, the instrument through which state Medicaid agencies report their actual program benefit costs and administrative expenses to CMS each quarter. The CMS-64 includes specific line items for total drug expenditures, federal (MDRP) rebate collections, and supplemental rebate collections, which I aggregated to annual state-level totals. CMS-64 data are available for fiscal years 2002–2024.

Data quality in the CMS-64 is uneven. Formatting and variable naming differ across fiscal years, some states have apparent reporting errors (e.g., implausibly negative supplemental rebate values), and the granularity of reported line items has changed over time. To validate the CMS-64-derived measures, I cross-referenced total drug spending against CMS State Drug Utilization Data (SDUD), which reports drug utilization and spending from a separate administrative source. Discrepancies were flagged and investigated case by case.

The resulting panel contains 1,150 state-year observations ($50 \text{ states} \times 23 \text{ years}$), with 39 observations (3.4%) missing the supplemental rebate outcome. The mean supplemental rebate percentage across all state-years is 2.8%, with substantial variation, from -8.1% to 28.4% . Six

state-years have negative supplemental rebate values. Four are negligible in magnitude (effectively zero), while two, North Carolina in 2013 (−8.1%) and Virginia in 2012 (−3.0%), likely reflect timing or reporting artifacts. These observations are retained in the analytic sample without adjustment; excluding them does not materially affect the results.

3.2 Pool Membership

The key independent variable is pool membership status: whether a state participates in SSDC, NMPI, TOP\$, or no pool in a given year. No pre-existing public dataset documents the full history of pool participation for all 50 states, so I constructed this dataset through three complementary approaches.

First, I systematically scraped state Medicaid agency websites for information on supplemental rebate agreements, PDL vendors, and pool participation announcements. Second, I contacted each state's official policy, rebate, and technical contacts for the MDRP and drug rebate programs by phone and email to confirm pool participation dates and any concurrent policy changes (e.g., PDL vendor changes, carve-in/carve-out transitions, uniform PDL implementation). Third, for states that did not respond or whose reported data were inconsistent with CMS-64 figures, I submitted FOIA or Public Records Requests to obtain supplemental rebate documentation.

I supplemented these primary data with expert interviews to provide context on pool structures and operations, and with secondary sources including KFF state surveys (KFF, 2020; KFF/HMA, 2020), NASHP policy analyses (Horvath, 2019), and OIG reports (OIG, 2014).

Pool indicators are mutually exclusive within a state-year: each state is classified as belonging to exactly one of SSDC, NMPI, TOP\$, or no pool. In 2002, the first year of the panel, all 50 states

were unaffiliated, as no pools had been established yet. By 2024, 15 states participated in SSDC, 11 in NMPI, 7 in TOP\$, and 17 in no pool.

3.3 SSDC Adoption Timing

The identification strategy relies on the staggered adoption of SSDC across states. Fifteen states adopted SSDC between 2006 and 2024, in ten distinct cohorts:

Cohort	States
2006	Iowa, Maine, Vermont
2007	Oregon, Utah, Wyoming
2011	North Dakota
2014	Mississippi
2015	West Virginia
2016	Delaware, Oklahoma
2017	Ohio
2020	Pennsylvania
2022	South Dakota
2024	Kentucky

The staggered timing and multiple adoption cohorts provide the variation needed for the difference-in-differences design. Early adopters (2006–2007) were predominantly smaller, rural states. Later adopters include larger states (Ohio, Pennsylvania) and states transitioning from other pools, which provides heterogeneity in pre-treatment conditions.

3.4 Summary Statistics

Table 1: Descriptive Statistics by Pool Membership Status

	No Pool	SSDC	NMPI	TOP\$	All States
Mean supplemen tal rebate (%)	1.93	5.15	2.83	3.33	2.83
Median supplemen tal rebate (%)	0.26	4.72	1.99	3.52	1.88
SD (%)	2.92	4.38	3.39	2.49	3.43
Min (%)	−2.98	0.00	−8.11	−0.08	−8.11
Max (%)	21.83	28.40	28.24	12.35	28.40
State-year	555	180	219	157	1,111

	No Pool	SSDC	NMPI	TOP\$	All States
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Note: The supplemental rebate percentage is supplemental rebate dollars divided by total drug expenditures, from the CMS-64. Negative values likely reflect reporting artifacts. State-years with a missing supplemental rebate outcome are excluded.

SSDC member state-years have substantially higher supplemental rebate percentages than all other categories. The gap between SSDC and no-pool states is around 3.2 percentage points at the mean and 4.5 percentage points at the median. NMPI and TOP\$ state-years are closer to non-pool states, particularly at the mean.

4. Empirical Strategy

Identification. The goal is to estimate the causal effect of SSDC adoption on supplemental rebate capture. The key identification challenge is that states selecting into SSDC may differ systematically from non-adopters in ways correlated with rebate performance. For example, states with stronger pharmacy programs or more political will to negotiate aggressively may be more likely to join SSDC and to independently achieve higher rebates. I address this using a staggered-adoption difference-in-differences design. In all specifications, states adopting SSDC are considered as the “treated” group, while comparison states serve as “untreated” controls. The core identifying assumption is that, absent SSDC adoption, treated states would have followed the same rebate trajectory as comparison states (parallel trends). I test this assumption using event-study specifications that allow examination of pre-adoption dynamics.

Event-study specification. The primary specification is an event-study regression estimated via two-way fixed effects with high-dimensional fixed effects (reghdfe):

$$Y_{st} = \alpha_s + \gamma_t + \sum_{k \neq -1} \beta_k D_{st}^k + \varepsilon_{st}$$

where Y_{st} = the supplemental rebate percentage for state s in year t , α_s and γ_t = state and year fixed effects, D_{st}^k = an indicator equal to one when an ever-treated state is k years from its first year of SSDC participation, and ε_{st} represents an error term. The β_k coefficients trace out the dynamic treatment effect relative to $k = -1$ (one year before adoption), which serves as the omitted reference period. Standard errors are clustered at the state level. Event time is binned at $k = -10$ and $k = +10$ to absorb extreme leads and lags and to improve estimation stability. The pre-adoption coefficients test for differential pre-trends, and the post-adoption coefficients estimate the dynamic treatment effect.

Specifications and comparison groups. I estimate four specifications that vary the comparison group and the sample period:

1. **SSDC vs. all others (2002–2024):** The treatment group is SSDC-adopting states, and the comparison group includes all non-SSDC states (NMPI, TOP\$, and no-pool states). This is the most inclusive specification (N=1,111 state-years, 50 states).
2. **SSDC vs. all others (2010–2024) (*preferred*):** Same comparison group, but restricted to 2010 onward (N=720, 50 states). This addresses the concern that the pre-2010 period, when pools were newly established and supplemental rebates were nascent, may not be representative of steady-state pool effects.
3. **SSDC vs. no-pool states only (2002–2024) (*preferred*):** Drops NMPI and TOP\$ states from the comparison group and uses only states that never joined any pool as controls (N=735, 48 states). This provides a cleaner comparison if NMPI/TOP\$ participation itself has (possibly attenuated) effects on rebates.
4. **SSDC vs. NMPI/TOP\$ states only (2002–2024):** Drops no-pool states and compares SSDC adopters directly to states in the other two pools (N=556, 37 states). This

specification attempts to isolate the SSDC-specific effect relative to alternative pooling arrangements. However, as discussed below, significant pre-trend violations limit its causal interpretability.

Robustness and diagnostics. I conduct several robustness checks. First, I assess parallel pre-trends visually through the event-study plots and statistically through joint tests of the pre-period coefficients. Second, I examine sensitivity to the choice of comparison group, as described above. Lastly, I consider the implications of recent advances in the staggered difference-in-differences literature. Standard two-way fixed effects (TWFE) estimators can produce biased estimates when treatment effects differ across cohorts (Goodman-Bacon, 2021; de Chaisemartin and D’Haultfoeuille, 2020). Callaway and Sant’Anna (2021) proposed group-time average treatment effect estimators that avoid “forbidden comparisons” between early- and later-treated units, and Sun and Abraham (2021) developed an interaction-weighted estimator for event studies under the same concern. Roth (2022) cautioned that conventional pre-trends tests have low power and may exacerbate bias when passed.

The primary implementation uses two-way fixed effects with event-study dummies. As a robustness check, I also estimate Callaway-Sant’Anna (2021) group-time ATTs using doubly-robust methods, which avoid forbidden comparisons and provide a formal Wald pre-test of parallel trends (Section 5.8). Given the relatively small number of clusters, I supplement asymptotic cluster-robust inference with wild cluster bootstrap p-values (Cameron, Gelbach, and Miller, 2008). The comparison of the full-sample estimate (1.55 pp) to the post-2010 estimate (3.15 pp) is consistent with positive heterogeneity across adoption cohorts, which under TWFE would attenuate the full-sample estimate toward zero. The preferred post-2010 specification mitigates this concern by focusing on a more homogeneous treatment period.

***## 5. Results

5.1 Full Sample: SSDC vs. All Others (2002–2024)

The first specification compares SSDC-adopting states to all non-SSDC states over the full panel period ($N=1,111$). The event-study estimates (Figure 1, Panel A) show pre-adoption coefficients that are individually insignificant and close to zero, supporting the parallel trends assumption.

Post-adoption coefficients turn positive and grow over time, though they are imprecisely estimated in the early post-adoption years. Effects become statistically significant at year 6 (coefficient = 0.030, $p < 0.001$), year 7 (0.029, $p = 0.013$), and year 10 (0.027, $p = 0.040$). The average post-adoption effect across all post-periods is 1.6 percentage points. This average is attenuated by the inclusion of the pre-2010 period, when the pools were nascent and supplemental rebate programs were still developing.

5.2 Preferred Specification: SSDC vs. All Others (2010–2024)

Restricting the sample to 2010 onward ($N=720$) produces substantially larger and more precisely estimated effects (Figure 1, Panel B). Pre-adoption coefficients remain individually insignificant and close to zero. Post-adoption effects emerge at year 3 (0.025, $p = 0.022$) and become highly significant and sustained after that: 0.036 at year 4 ($p = 0.002$), 0.033 at year 5 ($p = 0.002$), 0.049 at year 6 ($p < 0.001$), 0.048 at year 7 ($p < 0.001$), 0.056 at year 8 ($p = 0.007$), 0.039 at year 9 ($p = 0.002$), and 0.057 at year 10 ($p < 0.001$). Eight of eleven post-adoption coefficients are significant at the 5% level. The average post-adoption effect is 3.15 percentage points, and effects reach 4.8–5.7 percentage points at years 7 through 10.

Given that the baseline mean supplemental rebate for non-pool states is 1.9%, the preferred estimate of 3.15 percentage points represents a near-tripling of supplemental rebate capture relative to non-participation.

5.3 Alternative Comparison Group: SSDC vs. No-Pool States Only

Dropping NMPI and TOP\$ states and using only non-pool states as controls (N=735, 48 states) produces results closely aligned with the preferred specification (Figure 1, Panel C). Pre-adoption coefficients are individually insignificant and close to zero. Post-adoption effects are significant starting at year 3 (0.024, $p = 0.019$), with 8 of 11 post-adoption coefficients significant at the 5% level. The average post-adoption effect is 3.05 percentage points, with effects reaching 4.4–5.2 percentage points at years 6 through 10. The consistency with the preferred specification suggests that the main results are not driven by NMPI/TOP\$ states experiencing their own (potentially attenuated) treatment effects.

5.4 SSDC vs. NMPI/TOP\$ States: Pre-Trend Violations

The fourth specification compares SSDC adopters directly to states in the other two pools (N=556, 37 states; Figure 1, Panel D). This specification shows significant pre-trend violations. Several pre-adoption coefficients are negative and statistically significant (-0.065 at year -9 , $p = 0.020$; -0.068 at year -8 , $p = 0.020$; -0.066 at year -7 , $p = 0.022$; -0.068 at year -6 , $p = 0.012$), indicating that SSDC-adopting states had systematically *lower* supplemental rebate percentages than NMPI/TOP\$ states in the years before adoption. The average post-adoption effect is -1.6 percentage points, and no post-adoption coefficients are significant at the 5% level.

These pre-trend violations preclude a causal interpretation of this specification. However, they are informative in two respects. First, they suggest that the states joining SSDC were not the states already performing well on supplemental rebates. Rather, they may have been states dissatisfied with their rebate performance and looking for a more effective pooling arrangement. This selection pattern would, if anything, bias the preferred specifications against finding positive SSDC effects, which reinforces the credibility of the results. Second, the fact that SSDC

and NMPI/TOP\$ states were on fundamentally different rebate trajectories before SSDC adoption supports using no-pool states, rather than other-pool states, as the primary comparison group.

5.5 Results Across Specifications

Table 2 summarizes the results across all four specifications.

Table 2: Summary of Results Across Specifications

Specification	N	Avg Post Effect (pp)	Sig Coefs (p<.05)	R ²	Within R ²
SSDC vs All (2002–2024)	1,111	1.55	3/11	0.504	0.076
SSDC vs All (2010–2024)	720	3.15	8/11	0.688	0.177
SSDC vs No Pool (2002–2024)	735	3.05	8/11	0.588	0.111
SSDC vs NMPI/TOP\$ (2002–2024)	556	–1.56	0/11	0.587	0.108

Note: Bold indicates the preferred specifications. “Avg Post Effect” is the mean of all post-adoption coefficients (years 0 through 10). “Sig Coefs” reports the number of post-adoption coefficients significant at the 5% level.

5.6 Dynamic Treatment Effects and Cohort Heterogeneity

Across the preferred specifications, the event-study coefficients show a clear dynamic pattern: effects are small and insignificant in the first two post-adoption years, emerge at year 3, and grow steadily through year 10. This ramp-up makes sense given how supplemental rebate agreements actually get negotiated. Agreements are struck drug class by drug class, and the full benefit of pool membership accrues as the state progressively renegotiates its PDL under the new pooling arrangement.

The comparison between the full-sample and post-2010 specifications provides indirect evidence of cohort heterogeneity. The full-sample estimate (1.55 pp) is substantially attenuated relative to the post-2010 estimate (3.15 pp), consistent with later SSDC adopters seeing larger gains. This pattern suggests that SSDC's negotiating effectiveness has improved over time, potentially reflecting accumulated bargaining expertise, a larger pool of covered lives, and the proliferation of branded specialty drugs with high therapeutic substitutability that create stronger PDL leverage.

It is important to note that the number of states contributing to each event-time coefficient declines at longer horizons. At year 0, all 15 treated states contribute; by year 8, 11 states remain; and by year 10, only 8 states (the 2006–2007 and 2011–2014 cohorts) contribute. The large point estimates at years 8–10 should therefore be interpreted with caution, as they reflect the experience of predominantly early-adopting, smaller states.

5.7 Parallel Trends

Event-study plots for the two preferred specifications show pre-adoption coefficients that are individually insignificant and close to zero. In the post-2010 specification, the largest pre-period coefficient is -0.016 at year -6 ($p = 0.194$); in the no-pool comparison, the largest is 0.012 at year -6 ($p = 0.365$). There is no visual or statistical evidence of differential pre-trends in either preferred specification. Conventional pre-trends tests have limited power (Roth, 2022). However, the absence of pre-trend divergence, combined with the sharp onset of effects beginning at year 3 and growing steadily after that, supports a causal interpretation.

5.8 Heterogeneity-Robust Estimates and Inference

Callaway-Sant'Anna group-time ATT estimates confirm the main findings. In the preferred post-2010 specification, the CS overall ATT is 3.15 percentage points ($SE = 0.66$ pp, 95% CI:

1.86 to 4.44), nearly identical to the TWFE estimate. Restricting the no-pool comparison to 2010 onward similarly yields a significant CS ATT of 2.98 percentage points (SE = 0.71 pp, 95% CI: 1.59 to 4.37). Over the full 2002–2024 panel, the no-pool CS estimate is attenuated (2.21 pp) and not statistically significant, consistent with the TWFE finding that the pre-2010 data dilute the signal.

The CS event-study aggregations show near-zero pre-period estimates and a clear ramp-up beginning at year 3, closely matching the TWFE dynamic pattern. Manual Wald tests of the pre-period coefficients (assuming independence) fail to reject parallel trends for the full-panel no-pool specification ($W = 23.92$, $p = 0.297$) but reject for the post-2010 specifications. The rejections are driven by a single distant pre-period (year -13); the near-treatment pre-periods (years -1 through -5) are individually insignificant across all specifications.

Wild cluster bootstrap inference yields a p-value of 0.032 for the no-pool specification and 0.10 for the all-states post-2010 specification. The marginal bootstrap p-value for the preferred specification reflects how conservative bootstrap inference is with 49 clusters and a binary treatment; the result remains significant at the 10% level under this more demanding procedure.

6. Discussion

6.1 Interpretation of Findings

States that joined SSDC saw large gains in supplemental rebate capture. In the preferred specifications, SSDC participation increased supplemental rebate capture by around 3.1 percentage points annually, with effects reaching 4.8–5.7 percentage points at longer post-adoption horizons. Given a baseline supplemental rebate of 1.9% for non-pool states, this represents a near-tripling of rebate capture. The gains arrived through the SSDC rather than

through pooling generally, so interstate purchasing pool participation is not uniformly effective in increasing Medicaid supplemental rebates, and features beyond pooled population size, such as negotiating approach and administrative methodology, may be the key drivers.

The dynamic pattern of the effects is notable. The impact is small and insignificant in the first two years after adoption, emerges at year 3, and grows steadily through year 10. This trajectory aligns with the institutional reality that supplemental rebate agreements are negotiated drug class by drug class, and that the full benefit of pool membership accrues as the state progressively renegotiates its PDL under the new pooling arrangement. The gradual onset also makes it less likely that the effects reflect a one-time selection event or a concurrent policy change, which would typically produce a discrete jump at the adoption date.

The pre-trend violations in the SSDC-versus-NMPI/TOP\$ specification merit careful discussion. The significant negative pre-adoption coefficients indicate that future SSDC adopters had systematically lower supplemental rebate percentages than NMPI/TOP\$ states in the years before joining SSDC. This pattern is inconsistent with a story in which SSDC states were already high-performing and would have continued to outperform absent adoption. Rather, it suggests negative selection – states that joined SSDC may have done so precisely because their existing rebate capture was inadequate. If this selection pattern extends to the preferred specifications (where SSDC is compared to no-pool or all non-SSDC states), it would bias the estimates against finding positive effects. This makes the significant positive results in those specifications more credible, not less.

The pre-trend violations also mean that a direct causal comparison of SSDC to NMPI/TOP\$ is not feasible with the current design. The two groups of states differ in unobserved ways that correlate with their rebate trajectories. Future work with richer covariates (e.g., Medicaid

enrollment, pharmacy benefit structure, PDL vendor identity) might support a conditional parallel trends assumption, or alternative estimators (synthetic control, matching) could be explored.

SSDC's distinctive structural feature, that member states contract individually with manufacturers using state-specific SRAs while leveraging the consortium's collective purchasing power, may explain its differential effectiveness. In NMPI and TOP\$, the pool drives the negotiation and owns the contract paper, and individual states then decide which negotiated drugs to include on their PDLs. In SSDC, the state drives the negotiation and owns the paper, with the consortium providing leverage through pooled lives and a coordinated negotiating approach. This model may be more effective because it preserves state PDL autonomy and ownership of the negotiating relationship, which creates stronger incentives for aggressive negotiation and tighter alignment between the state's therapeutic priorities and its rebate strategy.

6.2 Policy Implications

These findings have a few main policy implications.

First, states considering joining a purchasing pool to increase supplemental rebates should not assume that pool membership alone will yield gains. The choice of pool, and specifically the pool's contracting structure, negotiating methodology, and administrative approach, is a first-order determinant of effectiveness. States should evaluate pool options based on these design features rather than on member count or geographic coverage alone.

Second, broader adoption of SSDC's methods could generate substantial system-wide savings. A rough extrapolation illustrates the potential magnitude. Fifteen states currently participate in SSDC, accounting for around \$20 billion of the \$80 billion in gross Medicaid drug spending. Applying the estimated 3.1-percentage-point effect to the remaining \$60 billion in non-SSDC

drug spending yields an upper bound of around \$1.9 billion annually. However, this is a mechanical extrapolation that assumes the estimated effect would translate fully at scale. If the effect were attenuated by 50 percent due to general equilibrium responses, such as manufacturer pricing adjustments to widespread SSDC-style contracting or diminished bargaining leverage from a larger consortium, the implied savings would be around \$0.9 billion. A plausible range is therefore \$1–2 billion annually. (A cruder calculation applying 3.1 percentage points to the full \$80 billion yields \$2.5 billion, but this overstates the addressable spending base and ignores equilibrium effects.) Even at the lower bound, the magnitude suggests that pool design reform is a high-return policy lever.

Lastly, improved transparency around pool participation, rebate outcomes, and related policy changes is essential for rigorous evaluation and evidence-based decision-making. The fact that constructing a basic panel of pool membership required web scraping, cold-calling state officials, and filing FOIA requests says a lot about the current opacity of this policy area. CMS or MACPAC could substantially improve the evidence base by requiring states to report pool membership and rebate outcomes in a standardized, publicly accessible format.

6.3 Relation to Prior Literature

This paper fills a gap identified across several literatures. The Medicaid pharmacy benefit literature has examined managed care's effects on drug spending (Dranove, Ody, and Starc, 2021) and PDL mandates' effects on utilization and costs (Munshi et al., 2018), but it has not examined the role of interstate purchasing pools. The pharmaceutical procurement literature has found that pooled procurement reduces prices in international settings (Dubois, de Bettignies, and Sampaio, 2021) and has identified institutional design as a key determinant of success (Seidman and Atun, 2022), but these predictions have not been tested in the U.S. Medicaid

context. The theoretical literature on formulary-rebate contracting (Ho and Lee, 2024) and buyer-side market power (Garthwaite and Scott Morton, 2017) predicts that pooling should increase rebates through enhanced bargaining leverage, but this prediction had not been empirically validated.

The finding that SSDC's individual-contracting-with-pooled-leverage model outperforms the collective contracting models is consistent with Seidman and Atun's (2022) identification of governance structure and member autonomy as key success factors in pooled procurement.

6.4 Limitations

My study design has several strengths, including a new 50-state panel built from primary sources, a staggered-adoption quasi-experimental design with 10 cohorts and 15 treated states, and an outcome validated against a second administrative data source. Potential limitations include measurement error. The supplemental rebate outcome is calculated from CMS-64 data, which has known quality issues. To address this, I 1) cross-validated the CMS-64 measures against SDUD, and 2) investigated anomalies through direct state outreach, but some measurement error is inevitable. To the extent that the measurement error is classical (random), it would attenuate the estimates toward zero and make the findings conservative. However, if reporting quality differs systematically between SSDC and non-SSDC states, the bias could go in either direction.

The identification strategy also relies on the parallel trends assumption – that absent SSDC adoption, treated states would have followed the same rebate trajectory as comparison states. The event-study plots support this assumption, but it is fundamentally untestable. States that select into SSDC may be responding to unobserved shocks (e.g., leadership changes, budget pressures) that independently affect rebate performance. I do not anticipate this driving the results, as the

staggered timing of adoption across 10 cohorts and 15 states makes it unlikely that all adopters experienced the same unobserved shock at their respective adoption dates.

The standard TWFE event-study specification may produce biased estimates under treatment effect heterogeneity (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfoeuille, 2020). The finding that later adopters experience larger effects is directionally consistent with positive heterogeneity, which would bias the TWFE estimate toward zero. As a robustness check, I implement the Callaway and Sant'Anna (2021) group-time ATT estimator, which avoids “forbidden comparisons” between early- and later-treated units. The CS estimates, reported alongside a formal Wald pre-test of parallel trends and wild cluster bootstrap p-values, provide a heterogeneity-robust benchmark for the TWFE results.

I also considered confounding from concurrent policy changes, which I cannot fully disentangle from the effect of SSDC membership, particularly uniform PDL adoption. The six highest-performing SSDC states (Delaware, Oklahoma, Vermont, Wyoming, West Virginia, and Maine) all maintain uniform PDLs that apply across both fee-for-service and managed care. If states tend to adopt uniform PDLs around the same time they join SSDC, the estimated effect may partly reflect the benefit of consolidated formulary control rather than the pool itself. Comprehensive state-level time-series data on uniform PDL adoption, pharmacy carve-in/carve-out status, and PDL vendor identity do not currently exist in any public source, which precludes including these variables as time-varying controls. My state outreach documented some of these transitions qualitatively, and the event-study's gradual post-adoption ramp-up (rather than a discrete jump coinciding with a single policy change) is inconsistent with a pure confounding story. However, residual confounding from correlated pharmacy benefit reforms cannot be ruled out. Future work

assembling a systematic panel of pharmacy benefit structure across all 50 states would substantially strengthen the identification strategy.

The analysis also treats pool membership as an absorbing state, but some states have exited pools or switched between pools over the study period. To the extent that these transitions are informative, a more granular treatment definition could improve precision.

Lastly, the results may not generalize to a counterfactual world in which all states adopt SSDC. If widespread adoption diminished SSDC's bargaining leverage (e.g., by reducing the threat of exclusion from a smaller pool) or triggered manufacturer pricing responses, the realized savings could be smaller than the state-level estimates suggest.

7. Conclusion

This paper provides the first quasi-experimental evidence on the effectiveness of Medicaid interstate drug purchasing pools, a policy mechanism adopted by two-thirds of U.S. states but never rigorously evaluated. Using a new state-level panel dataset and staggered-adoption difference-in-differences methods, I find that SSDC participation increases supplemental rebate capture by around 3.1 percentage points in the preferred specifications, with effects reaching 4.8–5.7 percentage points at longer post-adoption horizons. A direct comparison of SSDC to NMPI/TOP\$ states is confounded by pre-trend violations. However, the pattern of negative selection (future SSDC adopters had lower baseline rebates) reinforces the credibility of the positive findings in the preferred specifications.

These findings are directly relevant for state policymakers working on pharmaceutical cost containment. With Medicaid drug spending continuing to rise, supplemental rebates are one of the few levers states directly control. The evidence presented here suggests that the design of the

negotiating arrangement matters at least as much as the scale of the purchasing pool. States seeking to maximize supplemental rebate capture should examine SSDC's individual-contracting model as a potential template, and CMS should improve transparency around pool participation and rebate outcomes to enable ongoing evaluation.

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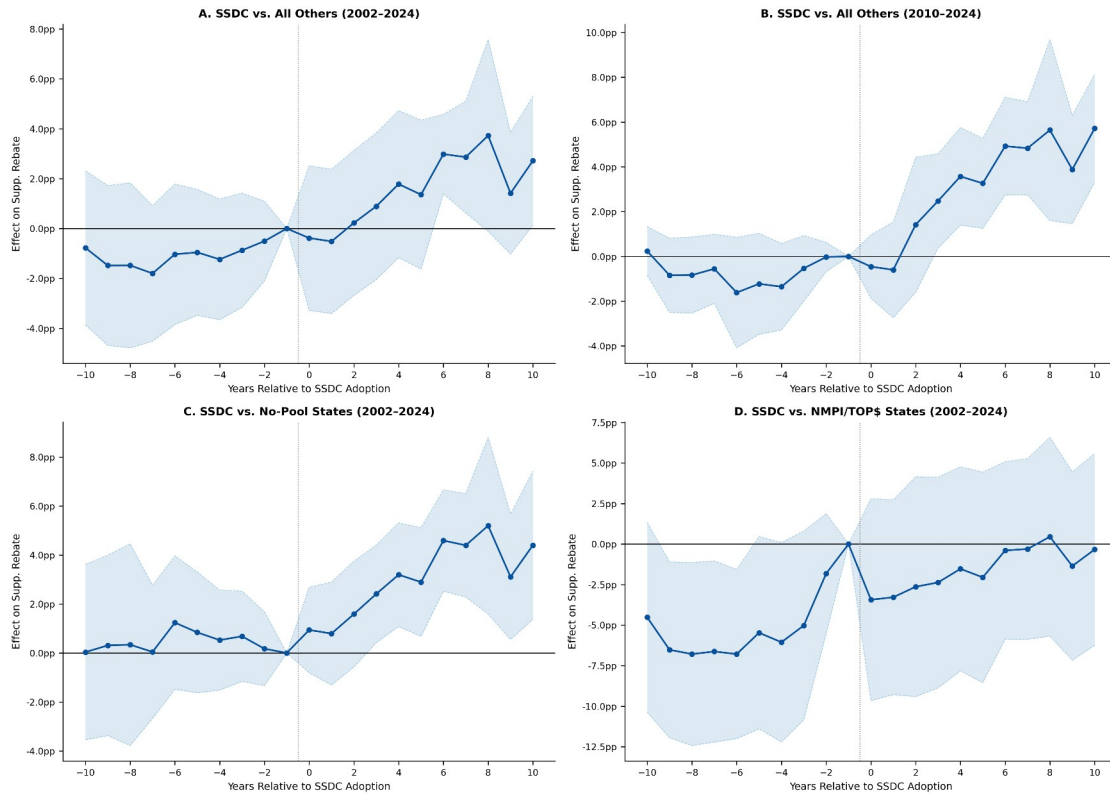
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Exhibits

Figure 1. Event-study estimates of SSDC adoption on supplemental rebate capture, by specification



Note: Each panel plots event-study coefficients for the effect of SSDC adoption on the supplemental rebate percentage, relative to the year before adoption (year -1). Shaded regions are 95% confidence intervals based on standard errors clustered at the state level. Panel A: SSDC vs. all other states, 2002–2024. Panel B: SSDC vs. all other states, 2010–2024 (preferred). Panel C: SSDC vs. no-pool states, 2002–2024 (preferred). Panel D: SSDC vs. NMPI/TOP\$ states, 2002–2024; this specification shows significant pre-trend violations and is not interpreted causally.